

Hochschild Homology in the Boutet de Monvel Calculus

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Abstract

Let X be a compact connected n -manifold with boundary, $n > 1$. We study the integer-order, type-zero Boutet de Monvel algebra modulo its residual ideal, with a specified topologically filtered Hochschild complex. The order spectral sequence has second page

$$E_k^2 \cong H^{2n-k}(S^*X \times S_{\sigma'}^1, S^*X|_{\partial X} \times S_{\sigma'}^1; \mathbb{C}).$$

The boundary term comes from the smooth Wiener–Hopf ideal and is essential: discarding it gives the wrong trace space. A strong convergence and E^2 -collapse theorem would turn the displayed formula into the full continuous Hochschild homology. We state that implication precisely, but do not attribute it to the groupoid theorem of Benamèur–Nistor, which does not cover the full Boutet de Monvel matrix calculus. Independently of collapse, the theorem of Fedosov–Golse–Leichtnam–Schrohe computes the Hausdorff degree-zero group: it is one-dimensional and is detected by the boundary noncommutative residue. Thus the manuscript separates the calculated page from the one remaining global homological input.

1 Scope and main statements

The expression *Hochschild homology of the Boutet de Monvel calculus* does not specify an algebra. The answer changes if one keeps smoothing operators, uses only order zero, admits arbitrary type, or changes the locally convex completion. We use the classical integer-order algebra of type zero, normalize its four entries with fixed order-reduction operators, and divide by its full residual ideal. This is the boundary analogue of the complete-symbol algebra on a closed manifold.

Set

$$Y = S^*X = (T^*X \setminus 0)/\mathbb{R}_+, \quad Z = Y|_{\partial X}.$$

Then Y is a compact manifold with boundary Z . The circle $S_{\sigma'}^1$ records integral symbol order; it is not an extra geometric circle in T^*X .

Theorem 1.1 (The computed page). *For the topologically filtered complete-symbol algebra defined in Section 2, the order spectral sequence has*

$$\boxed{E_k^2 \cong H^{2n-k}(Y \times S_{\sigma'}^1, Z \times S_{\sigma'}^1; \mathbb{C})}. \quad (1)$$

Equivalently,

$$E_k^2 \cong H^{2n-k}(Y, Z; \mathbb{C}) \oplus H^{2n-k-1}(Y, Z; \mathbb{C}).$$

Theorem 1.2 (Conditional full calculation). *Assume that the order spectral sequence is strongly convergent and collapses:*

$$E^2(\mathfrak{B}_{\text{cs}}) = E^\infty(\mathfrak{B}_{\text{cs}}) \quad \text{and} \quad E^\infty(\mathfrak{B}_{\text{cs}}) \implies \text{HH}_*^{\text{cont}}(\mathfrak{B}_{\text{cs}}). \quad (2)$$

Then the associated graded of the induced homology filtration is canonically given by the right side of (1). After choosing linear splittings,

$$\text{HH}_k^{\text{cont}}(\mathfrak{B}_{\text{cs}}) \cong H^{2n-k}(Y \times S_{\sigma'}^1, Z \times S_{\sigma'}^1; \mathbb{C})$$

noncanonically as vector spaces. In particular, it vanishes for $k > 2n$.

Theorem 1.3 (Unconditional degree zero). *For connected X , $n > 1$, the Hausdorffized continuous degree-zero homology is*

$$\overline{\text{HH}}_0^{\text{cont}}(\mathfrak{B}_{\text{cs}}) := \mathfrak{B}_{\text{cs}} / [\mathfrak{B}_{\text{cs}}, \mathfrak{B}_{\text{cs}}] \cong \mathbb{C}.$$

It is detected by the Fedosov–Golse–Leichtnam–Schrohe residue. This statement does not require (2). Without a separate closed-range theorem, the closure on the commutator space must be retained.

$$\begin{array}{ccccc}
\mathfrak{B}_{\text{cs}} & \xrightarrow{\text{order filtration}} & E^1 & \xrightarrow{\text{Poisson boundary}} & E^2 \\
& & & & \downarrow \text{convergence + collapse?} \\
\text{HH}_*^{\text{cont}}(\mathfrak{B}_{\text{cs}}) & \xleftarrow{\text{assembly}} & H^{2n-*}(Y \times S_\sigma^1, Z \times S_\sigma^1) & \xlongequal{\quad} & E^\infty
\end{array}$$

Figure 1: The symbol and Poisson steps compute E^2 . Passing from E^2 to the full homology requires the separately stated convergence-and-collapse input.

2 The algebra and topology

Let $E \rightarrow X$ and $F \rightarrow \partial X$ be complex vector bundles. An operator in Boutet de Monvel's calculus has the matrix form

$$A = \begin{pmatrix} P_+ + G & K \\ T & S \end{pmatrix} : C^\infty(X; E) \oplus C^\infty(\partial X; F) \longrightarrow C^\infty(X; E) \oplus C^\infty(\partial X; F). \quad (3)$$

Here P is a classical pseudodifferential operator on the double of X , its symbol satisfies the transmission condition, and $P_+ = r^+ P e^+$. The other entries are, respectively, a singular Green, potential, trace, and boundary pseudodifferential operator. We use type zero, which is closed under composition and contains parametrices after the usual order reductions [5, 7].

The four blocks have shifted natural orders. Choose elliptic scalar order reductions on X and ∂X . Conjugating by their diagonal matrix puts all blocks into one integer filtration. Different choices give filtration-preserving Morita equivalences. Denote the resulting algebra by

$$\mathcal{B}_0^{\mathbb{Z}}(X; E, F) = \bigcup_{m \in \mathbb{Z}} \mathcal{B}_0^m(X; E, F).$$

Its residual ideal $\mathcal{B}^{-\infty}$ consists of matrices whose entries have smooth kernels in the appropriate interior, boundary, trace, and potential spaces. Define

$$\mathfrak{B}_{\text{cs}}(X; E, F) = \mathcal{B}_0^{\mathbb{Z}}(X; E, F) / \mathcal{B}^{-\infty}(X; E, F). \quad (4)$$

Keeping $\mathcal{B}^{-\infty}$ produces a different extension problem and an additional ordinary operator trace.

We use the topologically filtered chain space of Benaméur–Nistor [3, Definition 1]. Write $F_{N,N}^N \mathfrak{B}_{\text{cs}}$ for their diagonal cofinal family: one first completes a finite symbol-order window in the decreasing order direction and only then lets the upper-order bound tend to infinity. In Hochschild degree q , set

$$C_q^{\text{tf}}(\mathfrak{B}_{\text{cs}}) = \varinjlim_N (F_{N,N}^N \mathfrak{B}_{\text{cs}})^{\widehat{\otimes}_{\pi} (q+1)}. \quad (5)$$

This notation packages the projective limits defining each completed filtration piece; it is not the tensor power of one order quotient. Multiplication is continuous on fixed pieces and sends a diagonal piece into a later diagonal piece, so the Hochschild boundary extends to (5). We denote its homology by $\text{HH}_*^{\text{cont}}(\mathfrak{B}_{\text{cs}})$.

Discussion. The topology in (5) is part of the problem. Taking one projective tensor power of the entire LF union need not make symbol composition jointly continuous. Formula (5) keeps each chain at finite upper order while allowing a complete classical asymptotic tail.

$$\begin{array}{ccc}
C^\infty(X; E) & \xrightarrow{P_+ + G} & C^\infty(X; E) & C^\infty(\partial X; F) & \xrightarrow{K} & C^\infty(X; E) \\
C^\infty(X; E) & \xrightarrow{T} & C^\infty(\partial X; F) & C^\infty(\partial X; F) & \xrightarrow{S} & C^\infty(\partial X; F)
\end{array}$$

Figure 2: The diagonal blocks act in the interior and on the boundary; the trace and potential blocks couple the two spaces. The singular Green correction belongs to the interior block.

3 The boundary symbol and the Wiener–Hopf ideal

There are two compatible full symbol systems. The interior symbol is the transmission symbol of P on $T^*X \setminus 0$. The boundary symbol is an operator-valued classical symbol on $T^*\partial X \setminus 0$, acting on

$$\mathcal{S}_+ \otimes E|_{\partial X} \oplus F, \quad \mathcal{S}_+ = \mathcal{S}(\overline{\mathbb{R}}_+).$$

In product coordinates its leading term is

$$a_\partial(x', \zeta') = \begin{pmatrix} p_+(x', 0, \zeta', D_n) + g(x', \zeta') & k(x', \zeta') \\ t(x', \zeta') & s(x', \zeta') \end{pmatrix}. \quad (6)$$

The Toeplitz part is determined by the boundary restriction of the interior symbol; g, k, t, s are independent boundary data.

Let $\mathcal{K}^\infty(\mathcal{S}_+)$ be the algebra of Schwartz-kernel compact operators. The smooth Wiener–Hopf fiber algebra has the linking ideal

$$\mathcal{J}^\infty = \begin{pmatrix} \mathcal{K}^\infty(\mathcal{S}_+) & \mathcal{S}_+ \\ \mathcal{S}'_+ & \mathbb{C} \end{pmatrix}. \quad (7)$$

Lemma 3.1 (Smooth Morita reduction). *The algebra $\mathcal{J}^\infty$ is nuclear and H -unital, and*

$$\mathrm{HH}_0^{\mathrm{cont}}(\mathcal{J}^\infty) \cong \mathbb{C}, \quad \mathrm{HH}_q^{\mathrm{cont}}(\mathcal{J}^\infty) = 0 \quad (q > 0).$$

The degree-zero map is the full linking trace

$$\begin{pmatrix} K & u \\ v & c \end{pmatrix} \longmapsto \mathrm{Tr}(K) + c.$$

Proof. Choose a rank-one projection e with Schwartz range. The corner $e\mathcal{J}^\infty e$ is $\mathbb{C}e$, and e is full. Schwartz nuclearity makes the completed finite-rank factorization strict, while the standard matrix-unit homotopy contracts the reduced bar complex. Thus $\mathcal{J}^\infty$ is smoothly Morita equivalent to $\mathbb{C}$. The same argument supplies the H -unitality required by continuous Hochschild excision [9]. $\square$

The quotient of the Wiener–Hopf algebra by $\mathcal{J}^\infty$ remembers the scalar normal symbol. The transmission condition identifies it with the restriction of the interior homogeneous symbol at Z . Therefore the order-graded Boutet de Monvel symbol algebra is a smooth pullback, not a direct sum.

$$\begin{array}{ccc}
\mathrm{gr} \, \mathfrak{B}_{\mathrm{cs}} & \xrightarrow{\quad} & C_{\mathrm{hom}}^{\infty}(T^*X \setminus 0) \\
\downarrow & & \downarrow i^* \\
C_{\mathrm{hom}}^{\infty}(T^*\partial X \setminus 0; \mathcal{W}^{\infty}) & \xrightarrow{\text{normal quotient}} & C_{\mathrm{hom}}^{\infty}(T^*X|_{\partial X} \setminus 0)
\end{array}$$

Figure 3: The interior and boundary symbols meet through the normal quotient. Hochschild excision turns this pullback into a relative, rather than direct-sum, form complex.

4 The first page

Filter a chain by the sum of the upper orders of its factors. Classical composition preserves the filtration. The leading product is the pointwise pullback product above. This constructs the order exact couple and its spectral sequence.

Use the mapping-cone model

$$\Omega^j(Y, Z) = \Omega^j(Y) \oplus \Omega^{j-1}(Z), \quad D(\alpha, \beta) = (d\alpha, i^*\alpha - d\beta). \quad (8)$$

Its cohomology is $H^j(Y, Z; \mathbb{C})$. The second component is a boundary transgression, not an independent boundary class.

Proposition 4.1 (Boundary HKR identification). *Continuous Hochschild excision for (7), followed by the HKR map on the scalar quotient, identifies the first page with the homogeneous mapping-cone complex for*

$$(T^*X \setminus 0, T^*X|_{\partial X} \setminus 0).$$

This identification is independent of collars and order reductions.

Proof. Work first in a product collar and trivialize E and F . The pullback of symbol algebras gives a short exact diagram whose boundary kernel is the Schwartz linking ideal. Lemma 3.1 and continuous excision replace that ideal by one scalar boundary complex. HKR identifies the scalar interior and normal quotients with homogeneous differential forms. The connecting map is restriction of the interior form to the boundary normal symbol. Thus the resulting total complex is (8), before radial reduction.

On overlaps, collar changes act by smooth Wiener–Hopf conjugations plus lower-order terms. Inner conjugations are chain homotopic to the identity, and lower-order terms do not alter this page. The local complexes are fine sheaves over X , so their local quasi-isomorphisms globalize. Order reductions change the model by a full smooth Morita corner. $\square$

5 The Poisson differential

For interior symbols a and b ,

$$\sigma_{m+m'-1}([A, B]) = \frac{1}{i} \{a, b\}.$$

Inserting this one-order-lower term in the Hochschild boundary and applying HKR gives the Brylinski differential

$$\delta = [\iota_{\pi}, d].$$

On the boundary, the regularized normal trace is not cyclic. Its commutator defect is the restriction of the interior Poisson term. This produces the second component $i^*\alpha - d\beta$ in (8); it is also the mechanism coupling the interior and boundary cochains in the boundary index cocycle [6, 4].

Let ω be the canonical symplectic form on $T^*X \setminus 0$ and $\mu = \omega^n/n!$. Symplectic duality

$$*_\omega \alpha = \iota_{\pi^\sharp \alpha} \mu$$

satisfies

$$*_\omega \delta = (-1)^{j+1} d *_\omega \quad \text{on } j\text{-forms.}$$

Stokes' formula supplies the boundary transgression, so this identity extends to the mapping cone. Hence the homology of the first differential is relative de Rham cohomology with complementary degree.

Choose a positive fiber norm r . For the Euler field $\mathcal{E} = r\partial_r$, Cartan's formula gives

$$d\iota_{\mathcal{E}} + \iota_{\mathcal{E}}d = \mathcal{L}_{\mathcal{E}}.$$

Thus every nonzero weight is contracted by $w^{-1}\iota_{\mathcal{E}}$. A weight-zero relative form splits uniquely as

$$(\alpha_0, \beta_0) + \frac{dr}{r} \wedge (\alpha_1, \beta_1).$$

Replacing dr/r by the positive generator of $H^1(S^1_\sigma)$ yields

$$H^j_{\text{hom}=0} \cong H^j(Y \times S^1_\sigma, Z \times S^1_\sigma).$$

This proves Theorem 1.1.

6 The remaining convergence-and-collapse problem

Benamou and Nistor compute complete symbols on differential groupoids and prove convergence and collapse under explicit anchor and local-triviality hypotheses [2, 3]. Their theorem applies to the b-groupoid algebra in the companion manuscript. It does not automatically apply to the matrix algebra (3).

The distinction is structural. Aastrup–Melo–Monthubert–Schrohe ask whether the full Boutet de Monvel algebra is a groupoid pseudodifferential algebra and construct a groupoid model for an important ideal, not an identification of the whole calculus [1]. The boundary symbol contains Toeplitz, Green, trace, and potential blocks; its pullback is not the scalar symbol algebra of a Lie algebroid.

Completeness on each fixed upper-order piece constructs the spectral sequence, but does not itself prove strong convergence after the strict inductive limit. The unbounded order direction may carry a derived-limit obstruction unless one proves a uniform regularity bound. Likewise, computing the Poisson differential does not rule out higher differentials.

A proof of (2) must:

1. prove strict continuous excision uniformly in symbol order and a regular convergence bound;
2. show that the Wiener–Hopf Morita homotopies preserve the complete order filtration;
3. construct permanent cycles for all relative E^2 -classes in a collar model;

4. globalize those cycles without losing asymptotic completeness; and
5. rule out all higher differentials.

These tasks fit the Benamèur–Nistor strategy, but none follows from Borel summation alone. Once (2) is proved, strong convergence identifies E^∞ with the associated graded of homology and linear splittings give Theorem 1.2.

7 Degree zero and the boundary residue

For a locally convex algebra,

$$\mathrm{HH}_0^{\mathrm{cont}}(A) = A / \mathrm{im}(b : C_1^{\mathrm{cont}}(A) \rightarrow A).$$

Its Hausdorffization is $A / \overline{[A, A]}$; its continuous dual is the space of continuous traces.

For A as in (3), write p_{-n} for the degree $-n$ interior symbol, $(\mathrm{tr}_n g)_{1-n}$ for the degree $1-n$ normal trace of the Green symbol, and s_{1-n} for the corresponding boundary symbol. With the standard cosphere densities, the Fedosov–Golse–Leichtnam–Schrohe residue is

$$\begin{aligned} \mathrm{Res}_{\mathrm{BdM}}(A) &= (2\pi)^{-n} \int_X \int_{S_x^* X} \mathrm{tr}_E p_{-n}(x, \xi) dS(\xi) dx \\ &\quad + (2\pi)^{1-n} \int_{\partial X} \int_{S_{x'}^* \partial X} [\mathrm{tr}_E (\mathrm{tr}_n g)_{1-n}(x', \xi') + \mathrm{tr}_F s_{1-n}(x', \xi')] dS(\xi') dx'. \end{aligned} \quad (9)$$

It is coordinate invariant and vanishes on commutators [6, 12, 8]. The interior and boundary integrals are not separately traces; their commutator defects cancel.

Under the connectedness and dimension hypotheses above, the FGLS trace theorem says that every continuous trace on the complete-symbol algebra is a multiple of (9). The residue is nonzero. Since the continuous dual separates points of a Hausdorff locally convex quotient, $\mathfrak{B}_{\mathrm{cs}} / \overline{[\mathfrak{B}_{\mathrm{cs}}, \mathfrak{B}_{\mathrm{cs}}]}$ is one-dimensional. This proves Theorem 1.3.

The top relative group gives the same count:

$$H^{2n}(Y \times S_\sigma^1, Z \times S_\sigma^1) \cong H^{2n-1}(Y, Z) \cong \mathbb{C}.$$

This is a consistency check, not a proof that the unclosed commutator range is closed.

8 Cyclic-cocycle check

The boundary index cocycle independently confirms the mapping-cone geometry. In the model of Boltachev–Savin, an interior cochain φ_{int} and a boundary regularized-trace cochain φ_∂ satisfy

$$b\varphi_{\mathrm{int}} = 0, \quad b\varphi_\partial + cB\varphi_{\mathrm{int}} = 0, \quad B\varphi_\partial = 0$$

for the normalization constant c fixed by the normal Fourier convention [4]. The middle identity is the cyclic counterpart of

$$D(\alpha, \beta) = (d\alpha, i^* \alpha - d\beta).$$

Thus the boundary cochain transgresses the restricted interior class. Treating it as an unrelated direct summand would incorrectly predict two degree-zero residue traces.

9 Checks and variants

9.1 Closed manifolds

If $\partial X = \emptyset$, then $Z = \emptyset$ and

$$E_k^2 \cong H^{2n-k}(S^*X \times S_\sigma^1; \mathbb{C}),$$

the Brylinski–Getzler complete-symbol calculation. In that setting the standard collapse theorem is available.

9.2 Boundary components

The pair (Y, Z) includes every boundary component. For connected X and $n > 1$, $H^{2n-1}(Y, Z) \cong \mathbb{C}$, so the residue is one trace, not one trace per boundary component. Lower relative groups retain the boundary topology.

9.3 Coefficient bundles

Stabilization by E and F does not change the filtered calculation. Locally this is matrix Morita invariance; globally the symbol modules form a full equivalence bimodule. The matrix traces in (9) implement the same Morita map.

9.4 Unreduced and order-zero algebras

The unreduced algebra belongs to the extension

$$0 \longrightarrow \mathcal{B}^{-\infty} \longrightarrow \mathcal{B}_0^{\mathbb{Z}} \longrightarrow \mathfrak{B}_{\text{cs}} \longrightarrow 0.$$

Its homology needs continuous excision and the homology of the chosen residual ideal. The ordinary kernel trace lives in that ideal and is absent from $\mathfrak{B}_{\text{cs}}$.

The C^* -closure of the order-zero calculus is the right setting for operator K -theory, but it is not (4). The work of Melo–Nest–Schick–Schrohe and Melo–Schrohe–Schick describes its K -theoretic symbol extension, not the Hochschild complex (5) [10, 11].

10 Conclusion

The order calculation gives relative cohomology of the cosphere pair, with the symbol-order circle, at E^2 . The boundary noncommutative residue verifies the top-degree prediction and computes Hausdorff degree zero without any collapse assumption. The sole remaining input for the all-degree formula is the explicitly stated Boutet de Monvel convergence-and-collapse theorem. This is narrower and more accurate than claiming that the existing groupoid complete-symbol theorem already covers the boundary matrix calculus.

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